

James Serrano

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EDUCATION

Yale University, New Haven, CT *Masters Science Computer Science*

SKILLS

AI, ML, Imitation and Reinforcement Learning, NLP, Python, Java, C, Rust, HTML, React, SQL, Racket, Bash Scripting, Assembly, Full Stack Web Development, Game Theory, LangChain, Gradle

WORK EXPERIENCE

Yale University New Haven, CT *Graduate Teaching Assistant Aug 2023 - May 2025*

- Delivered comprehensive instruction in Racket and Python, enhancing student comprehension through practical exercises.
- Designed and implemented an automated grading system, and coordinated scheduling of office hours of a team of 8 UTAs, which reduced student frustration, allowing for timely grading on midterms and final assessments while maintaining accuracy across 200+ student submissions.

George Mason University Fairfax, VA *Undergraduate Teaching Assistant Aug 2021 - May 2023*

- Facilitated engaging recitation sessions for over 60 students, increasing topic comprehension rates.
- Conducted weekly office hours, providing personalized support to more than 60 students.

PROJECT EXPERIENCE

Yale University, New Haven, CT *Project Lead on Robotic Imitation Learning AI, Jan 2024 - May 2024*

- Conducted comprehensive evaluations of over 15 imitation learning AI architectures, identifying the top 3 models that improved robotic task execution accuracy by 40%, leading to more adaptable and efficient automation processes.
- Developed and implemented testing protocols for diverse AI architectures, increasing testing throughput by 60% and reducing model validation time.
- Collaborated with cross-functional teams to integrate the most effective imitation learning models into robotic systems, resulting in an enhancement in task success rates across multiple operational scenarios.

Yale University, New Haven, CT *Project Lead on Facial Age Detector, Jan 2025 - May 2025*

- Created a toolset that utilized visual AI to estimate a person's age with an average slope of 0.997 and b value of 0.23

George Mason University, Fairfax, VA *Research Assistant on Experimental Topology, Jan 2021 - May 2023*

- Developed and implemented a Monte Carlo search algorithm for cubic shapes, resulting in a 100% increase in computational speed and enabling more efficient topology exploration across complex geometries.
- Conducted comprehensive statistical analyses to verify the accuracy of shape optimization results, achieving over 99.9% confidence in the model's reliability for diverse initial configurations.
- Adapted the algorithm to support other polyform shapes through rigorous modifications, facilitating broader applicability and uncovering critical insights into how initial shape variations influence topological outcomes.

UIUC, Urbana-Champaign, IL *Research Assistant on Graphene ML Analysis Jun 2022 - Aug 2022*

- Designed and implemented a novel computer vision pipeline to detect and quantify graphene quality, outperforming standard edge-detection and texture-analysis baselines
- Developed custom metrics for defect density, layer uniformity and surface continuity, enabling objective comparison across samples
- Benchmarked the method against established CV techniques such as Sobel filters, demonstrating improved sensitivity to micro-defects

LEADERSHIP EXPERIENCE

Yale University, New Haven, CT *Graduate Teaching Assistant Aug 2023 - May 2025*

- Coordinated and organized teaching assistants in course logistics, scheduling, and workload distribution, improving operational efficiency and response time to students.
- Served as a primary point of contact between students, undergraduate teaching assistants and the professor, streamlining communication and resolving scheduling and coverage issues.
- Managed TA assignments and task allocation to ensure balanced workloads and consistent support across discussion sections and office hours.

George Mason University, Fairfax, VA *Research Assistant Jan 2021 - May 2023*

- Onboarded and mentored new lab members on core algorithms, codebase structure, and research methodology, accelerating ramp-up time and productivity.
- Provided technical guidance on mathematical foundations and implementation details, helping junior members quickly contribute to ongoing projects.
- Acted as a go-to resource for debugging and conceptual questions, reducing friction and improving team velocity.